

Pareto Optimal Layouts for Adaptive Mixed Reality

Christoph Albert Johns
Aarhus University
Denmark
cajohns@cs.au.dk

Ken Pfeuffer
Aarhus University
Denmark
ken@cs.au.dk

João Marcelo Evangelista Belo
Aarhus University
Denmark
joabelo@cs.au.dk

Clemens Nylandsted Klokmose
Aarhus University
Denmark
clemens@cs.au.dk

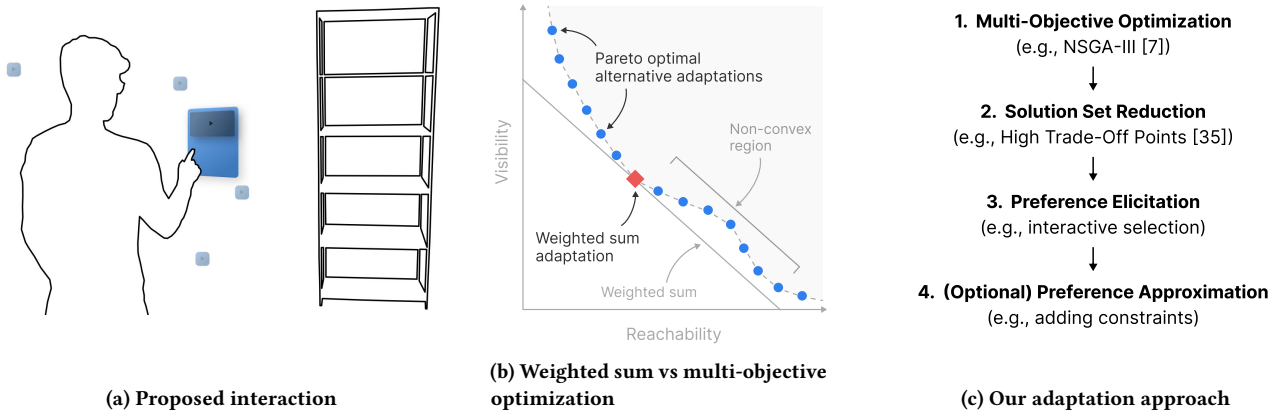


Figure 1: We present an approach for user interface layout adaptation in mixed reality applications that utilizes online multi-objective optimization. Multiple potential adaptations representing Pareto optimal trade-offs between pre-defined objectives are generated and presented to the user for selection. (a) Illustration of possible interactions with a mixed reality assembly application implementing our proposed approach (here, to place instructions on assembling furniture). (b) Illustrative comparison of an immediate adaptation found using weighted sum optimization (red) and our approach (blue). (c) Overview of our proposed method for optimization and preference elicitation.

ABSTRACT

Adaptive mixed reality applications adjust their user interfaces based on the context in which they are used to provide a smooth experience for different users and environments. This involves carefully positioning UI elements, which can be challenging due to the many possible placements and need to balance competing goals. Current approaches employ global criterion optimization methods like weighted sums, which can be difficult to use, inflexible, and might not find preferred solutions. This can prevent the adaptations from meeting end-user expectations. We suggest using online multi-objective optimization methods which generate a set of Pareto optimal adaptation proposals, giving users more control and adding flexibility to the computational decision-making. We explore the feasibility of our approach by generating adaptations

for a basic synthetic example and discuss relevant dimensions for a formal evaluation with end-users, including the choice of algorithm, decomposition technique, and objectives.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools.

KEYWORDS

multi-objective optimization, Pareto frontier, UI adaptation, mixed reality

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1 INTRODUCTION

Mixed reality (MR) applications blend digital and physical worlds, providing efficient access to digital information [11, 27]. Because

MR user interfaces (UIs) coexist in the physical world, they must adapt to dynamic contextual information such as user position, pose, task, and environment [27]. This includes adapting UI layouts to meet potentially conflicting objectives, such as visibility, input availability, and relevance [3]. An MR application supporting furniture assembly, for example, should consider the user's surroundings and physical comfort when positioning interactive instructions (see Figure 1a). If the user is working in a tight space, many placements may be occluded, unreachable, or uncomfortable, or may have to trade off some usability criteria for others (e.g., making an element easy to see at the cost of making it easy to reach). Consequently, reasonable placements must carefully balance objectives to find fair trade-offs.

To address this challenge, adaptation methods have been proposed that formalize UI placement as an optimization problem. Commonly, these formulations rely on a global criterion — an overall utility or cost score to be maximized or minimized — expressed as a weighted sum (e.g., [3, 11, 16, 27]). Global criteria suffer from various limitations regarding the solutions they can produce and the challenges they pose for designers and end-users [29]. Weighted sums, for example, cannot capture optimal points on non-convex regions of the Pareto frontier (i.e., the set of efficient trade-offs between conflicting objectives [29]; see Figure 1b) and require careful scaling and normalization of objectives [29]. In response to these limitations, alternative optimization methods have recently been proposed and employed in offline UI adaptation (e.g., [9, 23]). While these methods show promise for optimizing UIs at design time and in their respective task environments, they have not yet been applied to real-time adaptation of 3D UIs.

This paper proposes a novel approach to the real-time adaptation of MR interfaces (see Figure 1c). It formulates UI layout adaptation as an online multi-objective optimization problem without weights and uses state-of-the-art optimization methods to find its solutions. Users are presented with a meaningful representation of this set of optimal adaptation possibilities to pick from, in contrast to techniques that automatically choose a single solution. This can improve the perceived quality of adaptation as it allows users to choose adaptations that better meet their individual needs and preferences and provide users with the opportunity to learn about the trade-offs implicit in the adaptations. Furthermore, our approach can reduce the effort required to design and develop adaptive MR UIs by eliminating the need for static weights or arbitrary normalization often posed by current optimization-based adaptation systems. Lastly, it can capture potentially preferred UI adaptations that the commonly employed weighted sum-based optimization methods cannot (i.e., non-convex parts on the Pareto frontier) [29]. Overall, our approach promises to expose more of the design space of potential adaptations while reducing the set of presented solutions to those that are *optimal* or *efficient* according to some pre-defined criterion, improving user control compared to fully automated adaptation while reducing the cognitive load compared to fully manual adjustment.

We explore the feasibility of our proposed approach by adapting the position of a UI element in a basic 3D environment, comparing factors that might impact adaptations' usability, including the choice of objectives, algorithms, and decomposition techniques. By demonstrating the implications of our problem formulation, we

intend to open up several directions for future research, such as different optimization algorithms and techniques, alternative application domains, and various presentation and selection techniques.

2 RELATED WORK

Our work builds upon research on the adaptation of UI layouts in MR applications using global criterion optimization and previous efforts in multi-objective optimization of UIs outside the domain of MR. Global criterion optimization for adaptive MR refers to the real-time adaptation of MR interfaces using online optimization of a single scalarized utility or cost function.

2.1 Optimization of User Interfaces

Two important milestones for applying optimization techniques in UI design were the representation of design factors as decision variables in problem formulations and the modeling of design heuristics and psychological models in cost functions [33, sec. 4.2]. Pioneering work proposed novel objective functions and used existing optimization techniques to optimize keyboard layouts [8, 26, 42] and menu design [1, 22, 28, 30]. Since then, multi-objective optimization approaches have been applied to the design and offline adaptation of, for example, text entry methods (e.g., [5, 14, 21, 34, 36, 37]) and graphical UIs (GUIs) (e.g., [39]). A key work in this area was the multi-objective optimization method proposed by Smith et al. [36] which utilized a weighted sum-based scalarization with varying weights to generate Pareto optimal keyboard layouts. This approach has since been applied by, for example, [5] and [21] to generate keyboard layouts for constrained gesture typing and one-handed text entry respectively. In the domain of GUI optimization, SUPPLE [17–19] framed GUI design as a scalarized multi-objective optimization problem based on navigation costs and demonstrated the effectiveness of this approach in adapting to both device and usage characteristics using an equally-weighted sum [17]. Since SUPPLE, several studies have utilized similar approaches to, for example, optimize the layout of menu systems [2] or websites [39]. Recently, alternative multi-objective optimization methods have been utilized in the domain of keyboard layout optimization [23], computational design of physical input devices [24] and parameter search for 3D touch interaction [9]. These efforts employ more complex optimization methods (i.e., genetic algorithms [23] and Bayesian multi-objective optimization [9, 24]) to overcome the limitations of weighted sum approaches and find optimal configurations before deployment. Overall, multi-objective optimization of UIs has been dominated by weighted sum-based methods used at design time. As the recent efforts described above demonstrate, however, alternative approaches exist that aim to address some of the shortcomings of weighted sum-based optimization for UI adaptation. This paper contributes to this growing area of research by exploring the use of state-of-the-art multi-objective optimization techniques for online layout adaptation of MR UIs.

2.2 Global Criterion Optimization for Adaptive Mixed Reality

Several works have applied optimization approaches to adaptive layout problems in MR. These previous efforts are commonly concerned with optimizing UI elements' visibility (e.g., [16]) and spatial

coherence (e.g., [11, 32]), but have also considered other objectives such as ergonomics [15] and cognitive load [27]. Commonly, these optimization-based adaptation approaches treat layout adaptation as a single-objective optimization problem, combining multiple objectives for multiple elements into a single cost to be minimized or a single utility score to be maximized and returning a single solution. Most frequently, this global criterion takes the form of a static weighted sum (e.g., [3, 11, 16, 20, 27, 32, 41]), which can be optimized using commercial solvers for linear programs (e.g., [11, 27]) or using approximate techniques like simulated annealing (e.g., [3]). In contrast to these previous efforts, our approach treats layout adaptation as an online multi-objective optimization problem that returns a set of optimal adaptations. Consequently, our approach is not restricted to the same requirements as global criterion methods, for example regarding the formulation of static weights or normalization [29], and can explore novel interfaces that provide more flexibility to users, allowing them to choose from multiple solutions along the Pareto frontier. Notably, this does not implicate that adaptations cannot be fully automatic. Even with a Pareto set of solutions, the system can take initiative by picking one solution and showing its alternatives.

3 APPROACH

Our approach to real-time adaptation of MR interfaces uses multi-objective optimization to generate a varied set of optimal adaptations that users can choose from, positioning it between fully automated and fully manual adaptation. It can be invoked by a variety of triggers such as a threshold function or a regular interval [3]. Our approach can be divided into three major steps: Multi-Objective Optimization, Solution Set Reduction, and Preference Elicitation (see Figure 1c). Each of these steps reduces the set of potential adaptations until one is picked by the user through interaction or the set of solutions is reduced to a single adaptation. In an optional fourth step, the user's input can be used to inform an approximation of the latent preference function for future adaptations. Each of the four steps and their application to the online adaptation of layouts in MR is explained in further detail below.

3.1 Multi-Objective Optimization

Multi-objective optimization refers to finding the set of optimal solutions for a set of objective functions given some inequality and equality constraints [29]. It can, for example, be expressed using the following formulation:

$$\begin{aligned} \underset{\mathbf{x}}{\operatorname{argmin}} \mathbf{F}(\mathbf{x}) &= [F_1(\mathbf{x}), F_2(\mathbf{x}), \dots, F_k(\mathbf{x})]^T \\ \text{subject to } g_j(\mathbf{x}) &\leq 0 \text{ for } j = 1, 2, \dots, m \\ \text{and } h_l(\mathbf{x}) &= 0 \text{ for } l = 1, 2, \dots, e, \end{aligned} \quad (1)$$

with $\mathbf{F}$ as the set of k objective functions to be minimized using the design vector $\mathbf{x}$ as constrained by m inequality constraints g and e equality constraints h . For example, a problem setting such as the instructed assembly described in the introduction might involve the joint optimization of two objectives F_1 and F_2 that represent the expected neck and shoulder strain caused by a layout configuration $\mathbf{x}$ which is constrained by allowed placements expressed as inequalities g and equalities h (see Section 4).

Typically, these optimization problems do not yield a single globally optimal solution, but instead, a set of solutions that are optimal according to some chosen definition [29]. In our case, we utilize Pareto optimality as our definition of optimality. A point in the solution space is said to be Pareto optimal if no other feasible point exists that improves upon performance in one objective without degrading another [29]. Many multi-objective optimization methods exist that can generate Pareto optimal solutions (see, for example, [29] for an overview). To give one example, we may choose to employ a recent evolutionary algorithm called non-dominated sorting genetic algorithm III (NSGA-III) [7]. NSGA-III is a state-of-the-art many-objective extension and refinement of the multi-objective genetic algorithm NSGA-II [13]. It has been shown to effectively generate the Pareto frontier for a wide set of many-objective optimization problems as well as finding global optima for single-objective functions [12].

The result of the multi-objective optimization step of our approach is an even distribution of optimal adaptations along the Pareto frontier (see Figure 1b for an illustration). Applying this to our illustrative example, the multi-objective optimization step would yield a large set of optimal UI placements in the user's environment, representing a variety of trade-offs between the neck and shoulder ergonomics.

3.2 Solution Set Reduction

Taking the set of Pareto optimal solutions as input, the second step in our proposed pipeline deemed Solution Set Reduction aims to select a limited set of qualitatively different solutions from the Pareto frontier. Literature on multi-criteria decision-making has proposed a variety of methods to achieve this goal, differing in the specific qualities of their proposals (see, for example, [38] for an overview). We may choose, for example, to employ the Augmented Achievement Scalarization Function (AASF) described in [40] to identify a small set of extreme adaptations as well as a sample of compromise solutions as basis for selection and correction. AASF selects solutions based on their distance to a set of developer-defined reference points such as single-objective optima or specific targeted compromise solutions [40]. In our illustrative example, these points might include UI placements corresponding to the single-objective optima of both the neck strain and shoulder strain objectives and an equal trade-off solution. However, many alternatives exist that can be tailored to the specific application example. A few of these techniques are illustrated and discussed in Section 4.3.

3.3 Preference Elicitation

We refer to the activity of presenting the reduced set of Pareto optimal solutions to the user for interaction as Preference Elicitation. In this step, the user makes their preferences for certain solutions explicit by choosing which proposed adaptations to interact with. While formally an optional step since the previous Solution Set Reduction might yield only a single optimal solution (e.g., in the case of a single global optimum) or because the usage context might call for an immediate single adaptation (e.g., if the user is currently moving), we emphasize the promise of this step to improve user control and therefore detail it here. In our adaptive MR application, we may, for example, choose to represent the reduced set of Pareto

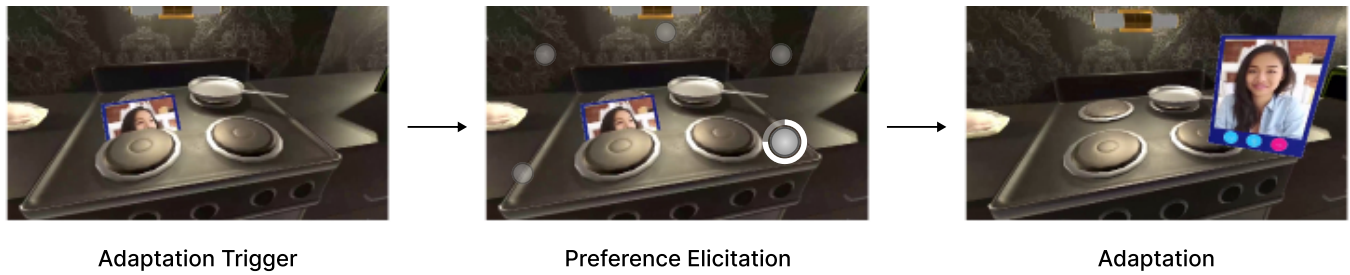


Figure 2: Mock-up of an implementation of our adaptation approach, adapted with permission from [3]. An adaptation is triggered (e.g., because of occlusion; left), the user preferences are queried (middle) and the UI element is moved to the selected position (right).

optimal placements using small grey spheres placed in the user’s environment to limit the visual load of the Pareto frontier presentation (see Figure 2 for a mock-up illustration). If the user selects one of these spheres by touching it for a pre-defined dwell time, a new instance of the UI element is animated to quickly fade in anchored at that position. The previous instance of the UI element is animated to fade out, allowing for a smooth transition between adaptations. Depending on the specific adaptation and application context, other methods of preference elicitation may be more appropriate. If the UI is particularly intricate, for example, designers may choose to represent the potential adaptations as floating 3D buttons (see Figure 1a) or, if the UI is rather simple, as semi-transparent duplicates of the original UI element (see Figure 3). Similarly, in the case of a multi-element UI, the user may need to preview the adapted position for each element in a proposal before making a selection. If elements need to be positioned outside the user’s reach, alternative selection techniques like raycast-based selection might be more appropriate. To give one final example, if a more automatic adaptation is desired (e.g., because an application is highly time-sensitive or otherwise occupies the end-user), an adaptation proposal may be chosen that acts as a default adaptation if the user does not provide any further input within a timeout period (e.g., by selecting a different proposal). Which presentation and selection format is appropriate for a particular application may highly depend on the specific usage domain and application characteristics.

3.4 (Optional) Preference Approximation

The final and optional step in our proposed approach is Preference Approximation. Interpreting the user’s selection from a set of presented solutions as an explicit statement of preference, this information can be used to inform an approximation of the user’s latent preference function. This approximation could then, in turn, be used to inform future runs of the multi-objective optimization, for example by adding further constraints (see, for instance, [19]) or by refining a model of the end-user preferences (cf. [25]). While our particular implementations do not take advantage of the user selection due to its effects on system complexity, a continuously updating preference function could be used to effectively improve later adaptations.

4 EXPLORATIVE SIMULATIONS

To validate the feasibility of our approach and explore how relevant factors affect usability, we implement an online optimization system for a UI consisting of a floating panel in a 3D indoor scene. Varying the objectives, algorithms, hyperparameter settings, and decomposition techniques, we explore how these factors affect the proposed adaptations. We create a Python-based multi-objective optimization solver for 3D layout problems using pymoo [6], that can be interfaced with MR adaptation tools such as AUIT [3]. All experiments are conducted on a MacBook Pro with an Apple M1 Pro CPU.

4.1 Experiment 1: Optimization Methods and Objective Formulations

In our first experiment, we compare the UI adaptations suggested by an equally weighted sum optimization with those derived by the NSGA-III [7] algorithm for a set of two bi-objective optimization problems to illustrate shortcomings of weighted-sum approaches and explore whether evolutionary algorithms can overcome them. Each of the two problem definitions aims to minimize a set of two objectives: the angle between the user’s line of sight and a UI element (neck angle, F_1) and an arm ergonomics metric that is either defined as bounded and continuous (F_2) or unbounded and discrete (F'_2). The bounded, continuous objective F_2 is chosen as the angle between the user’s resting arm and the UI element (arm angle; see Figure 3, *Continuous Objective*; cf. [15, 31]). The discrete, unbounded objective F'_2 is defined as a step-wise function of the Euclidean distance from the UI element’s position to the user’s current hand position (hand reachability). The zone immediately surrounding the current hand position is considered inconvenient and is therefore imbued with a high positive cost. The objective function then has a global minimum for the next closest zone and increases in equally sized steps for the zones beyond that. Such unbounded objectives are hard or even impossible to meaningfully normalize and scale, making them unsuitable for global criterion optimization methods such as weighted sums [29]. Both problems are subject to reachability constraints, formulated here as the distance from the element’s position to the user’s closest possible hand position given an outstretched arm, with unreachable positions considered a constraint violation. In practice, multi-objective UI optimization is likely dependent on a range of usability criteria and constraints [3].

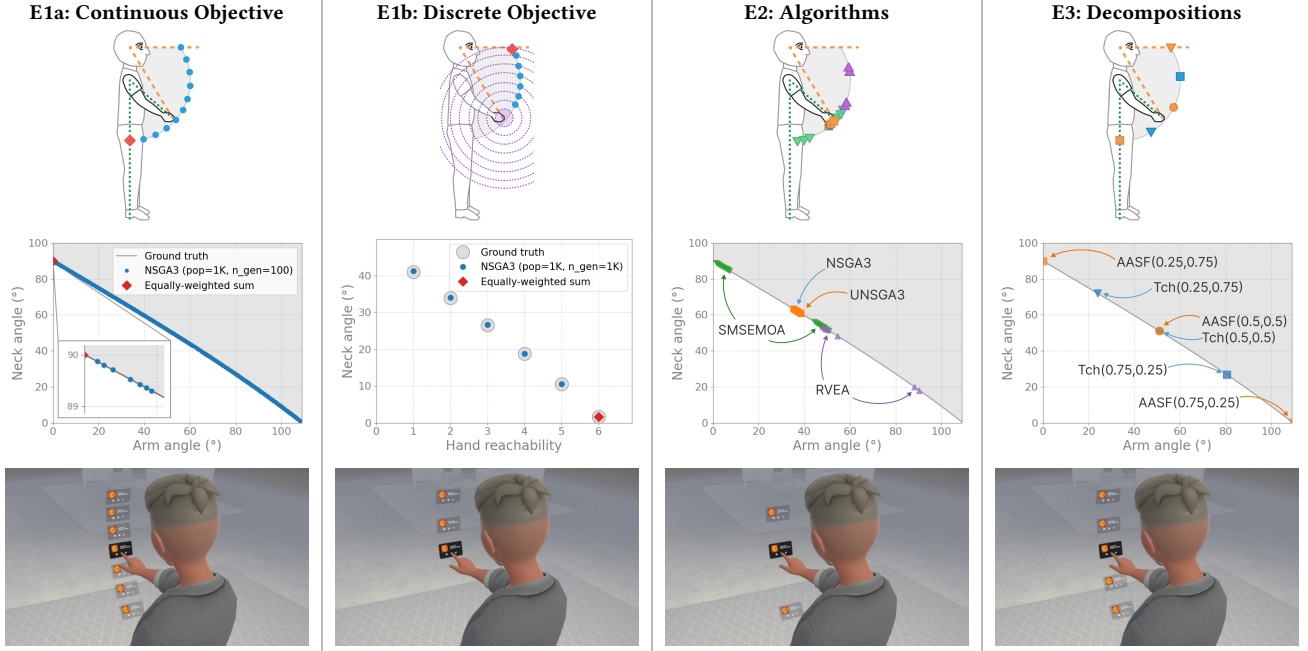


Figure 3: Results of experimental simulations exploring the effect of continuous and discrete objective formulations (left, middle left), algorithm (middle right), and decomposition technique (right) on adaptations derived via online multi-objective optimization. Each simulation is visualized as a 2D illustration of the UI adaptations (top), in a scatter plot containing the true Pareto frontier and the adaptations suggested by the solver (middle), and in a 3D scene featuring the UI (bottom).

For the sake of simplicity and to be able to validate our approach, we have limited our experiments to these problem definitions.

The equally weighted sum optimization does not yield a fair compromise between the two objectives for either scenario (see Figure 3, *Continuous Objective* and *Discrete Objective*). It, instead, produces an adaptation equivalent to a single-objective optimum due to the concave shape of the Pareto frontier. While this outcome may be viewed as a result of insufficient normalization, it is not straightforward how to normalize the objective functions in a way that would produce a fair compromise [29]. By contrast, the NSGA-III algorithm accurately approximates the true Pareto frontier — including the expected Pareto-optimal solution that equally balances both objectives — for both optimization problems, mirroring results from operations research literature (e.g., [7]). This suggests that optimization methods that return a representative approximation of the true Pareto frontier, like the NSGA-III algorithm, may form a more suitable basis for UI adaptation in MR applications than the common weighted sum-based optimization.

4.2 Experiment 2: Evolutionary Algorithms

In our second experiment, we compare the adaptations suggested by the NSGA-III algorithm with those derived by other common evolutionary algorithms (here, U-NSGA-III [35], RVEA [10], and SMS-EMOA [4]) for the optimization problem described in Experiment 1, *Continuous Objective*, to illustrate how the choice of multi-objective optimization algorithm can affect the proposed solutions. Given sufficiently large populations and sufficiently many generations,

all four algorithms are able to find a representative approximation of the true Pareto frontier. This suggests that the choice of multi-objective optimization algorithm might not be as important as the choice of objective functions and the number of generations and population size. To illustrate the effects of algorithm choice in an extreme scenario, however, we reduce the population size and generations to a minimum of 20 and 100, respectively, and compare the results of the four algorithms (see Figure 3, *Algorithms*). The results indicate that the choice of algorithm can have an effect on the suggested adaptations, but only under specific conditions.

4.3 Experiment 3: Decomposition Techniques

For our third and final experiment, we compare the adaptations suggested by the NSGA-III algorithm for different Solution Set Reduction (i.e., decomposition) techniques. These techniques can be used to reduce the set of Pareto optimal solutions to a smaller set (e.g., to present to users) or even a single solution (e.g., for an initial adaptation) and might therefore strongly affect usability. Here, we compare two common decomposition techniques that can incorporate static weights as statements about end-user preferences: Tchebicheff scalarization and AASF [40] (see Figure 3, *Decomposition*). For each of the techniques, we plot the suggested adaptations — given a sufficiently accurate approximation of the Pareto frontier (for example, as derived by an effective evolutionary algorithm) — with three different weight combinations: (1) $w_1 = 0.25$ and $w_2 = 0.75$, (2) $w_1 = w_2 = 0.5$, and (3) $w_1 = 0.75$ and $w_2 = 0.25$. The

results suggest that the choice and parameterization of the decomposition technique can have a significant effect on the suggested adaptations.

5 DISCUSSION

Our experiments indicate that multi-objective optimization algorithms like NSGA-III can find a range of Pareto-optimal solutions for various adaptation objectives, whereas weighted sum optimization can produce unexpected or undesired adaptations, depending on objective choice and scaling. Our results further suggest that the choice of algorithm and hyperparameter values can impact the distribution of generated potential adaptations and that the choice of decomposition technique used to reduce the set of presented adaptations can strongly influence system usability, whether it is used to select multiple adaptations to present to users or to choose a single immediate adaptation.

By posing the problem of real-time adaptation of MR UIs as an online multi-objective optimization problem, we have shown that it is possible to generate a set of Pareto optimal potential adaptations for MR UI layouts that can be presented to users for selection. This positions our proposed approach between fully automated techniques, for example, based on global criteria, and fully manual adaptations as present in many current MR systems. However, our research is still in its early stages and there is a lot of room for further investigation.

Presentation and Selection. It is unclear how the potential adaptations should be presented to users, how many solutions to present, and how users should be informed about the trade-offs between different objectives. It is also uncertain how users should select from the presented adaptations. We expect large benefits for system usability if the advantages of fully automated and fully manual adaptation can be combined in an adaptation technique. Techniques such as preference ordering (e.g., calculating distance to a utopia point [29]) may aid in this issue.

Cognitive Load. It is unclear how our approach affects cognitive load, including the presentation and selection techniques and the choice of algorithm, objective functions, and decomposition techniques. Our proposed interaction might increase cognitive load compared to current automatic single adaptations as the user can exert more control over the adaptation process at the cost of additional options and visual affordances. It could, however, also decrease overall cognitive load as the user might be able to select and interact with more usable adaptations. We currently suspect that our approach will decrease cognitive load as the user gains control over the adaptation. We expect, however, that the choice of presentation technique and the current usage context will have a strong influence on overall cognitive load and plan to investigate this effect with end-users.

Preference Approximation. How the user's latent preference function can best be approximated and integrated into the optimization procedure remains an open question. We suspect that multi-objective Bayesian optimization that uses information gain to guide solution set reduction could be a promising candidate, but many alternatives exist.

6 CONCLUSION AND FUTURE WORK

In this paper, we proposed a novel technique for real-time UI adaptation in MR applications. Our approach generates a set of Pareto-optimal potential adaptations that users can choose from, rather than relying on global criterion optimization methods that may not meet their expectations. We investigated and discussed factors that can impact adaptation usability, including objectives, algorithms, and decomposition techniques. Our experiments indicated that multi-objective optimization algorithms like NSGA-III can find a range of Pareto-optimal adaptations for various objectives and can be combined with various decomposition techniques to afford fully automated, semi-automated, or largely manual adaptation. We have discussed several opportunities for future research, including exploring the presentation and selection of adaptations and evaluating the approach with end-users. While this work has focused on adapting the position of a single UI element, we suspect that our approach can generalize to multiple-UI scenarios as well as to other problems beyond layout adaptation. These use cases will require new investigation of presentation and selection techniques and can offer a wealth of avenues for future research.

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